

# PRACTICAL NO:6

AIM:6. Decision Making under Uncertainty:

- Develop a decision tree model to make business decisions considering uncertainties and associated probabilities at each decision point.
- Apply the concept of expected value to evaluate different decision alternatives and select the optimal one.

## A) Decision Tree Model

```
"decision (a).py - C:/Users/info/OneDrive/Desktop/prac 6 decision/decision (a).py (3.14.0)"
File Edit Format Run Options Window Help
print("T110,SANJANA SAHU")
import matplotlib.pyplot as plt
from matplotlib.patches import FancyBboxPatch

# Create figure
fig, ax = plt.subplots(figsize=(12, 7))

ax.set_xlim(0, 12)
ax.set_ylim(0, 8)
ax.axis("off")

# Decision box
decision = FancyBboxPatch(
    (4.5, 6.5), 3, 1,
    boxstyle="round,pad=0.1",
    edgecolor="black",
    facecolor="lightblue"
)

ax.add_patch(decision)
ax.text(6, 7, "Business Decision", ha="center", va="center", fontsize=14)

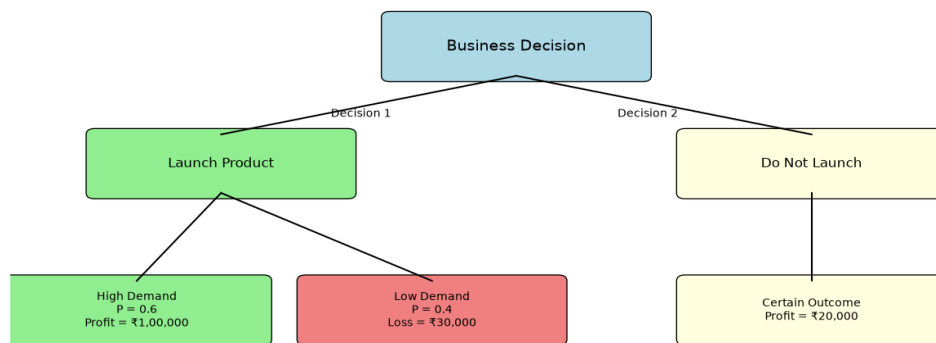
# Launch Product
launch = FancyBboxPatch(
    (1, 4.5), 3, 1,
    boxstyle="round,pad=0.1",
    edgecolor="black",
    facecolor="lightgreen"
)

ax.add_patch(launch)
ax.text(2.5, 5, "Launch Product", ha="center", va="center", fontsize=12)

# Do not launch
no_launch = FancyBboxPatch(
    (8, 4.5), 3, 1,
    boxstyle="round,pad=0.1",
    edgecolor="black",
    facecolor="lightyellow"
)

ax.add_patch(no_launch)
ax.text(9, 5, "Do Not Launch", ha="center", va="center", fontsize=12)
```

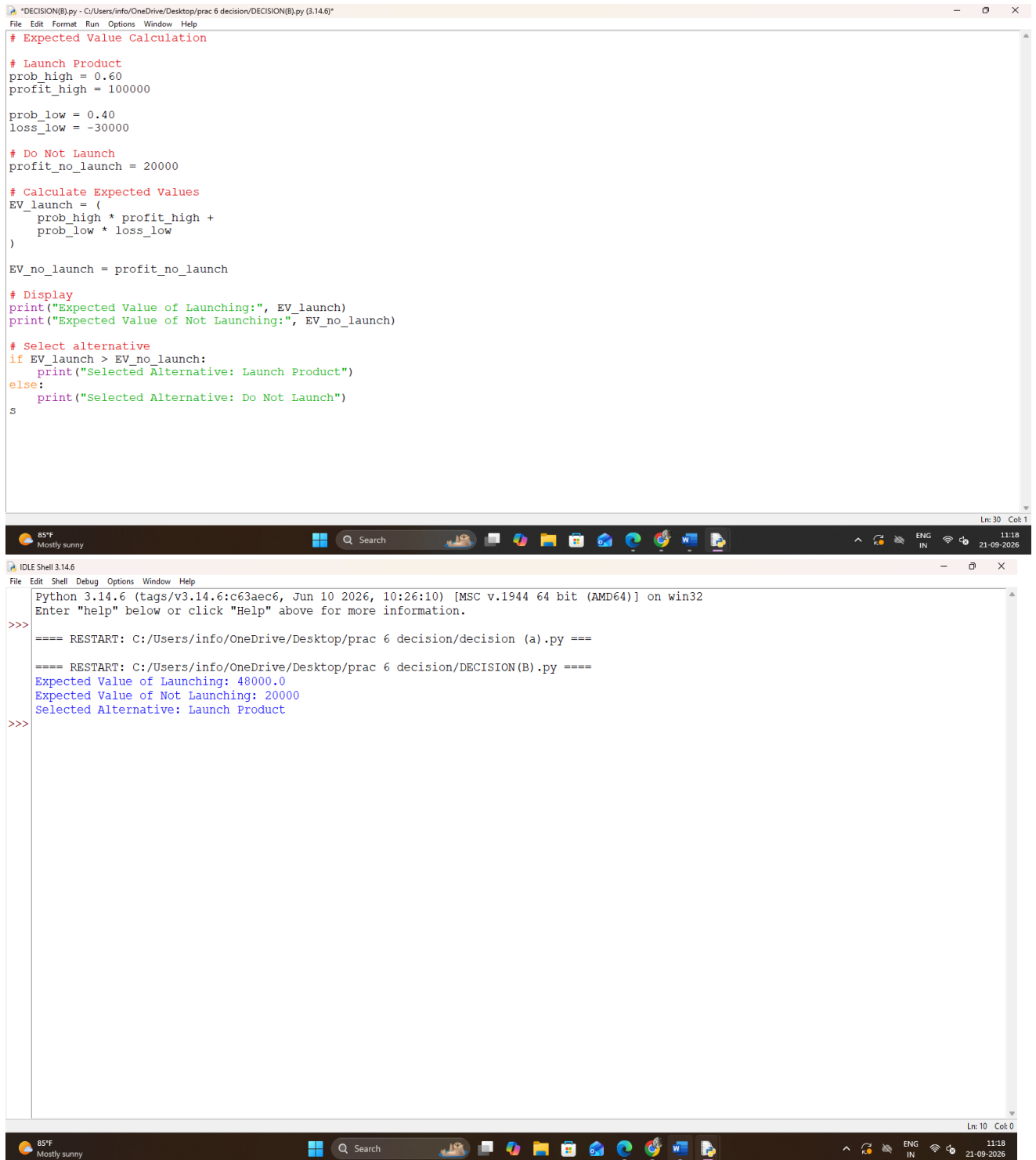
Decision Tree for Business Decision Making



## B) Expected Value Calculation

T110 SANJANA SAHU

# PRACTICAL NO:6



The image shows a screenshot of a Windows desktop with two windows open. The top window is a Python IDE titled "DECISION(B).py - C:/Users/info/OneDrive/Desktop/prac 6 decision/DECISION(B).py (3.14.6)". It contains a Python script for a decision analysis. The script calculates the Expected Value (EV) for launching a product versus not launching it, based on probabilities and profits/losses. The bottom window is an "IDLE Shell 3.14.6" showing the execution output of the script. The output indicates that the Expected Value of Launching is 48000.0, the Expected Value of Not Launching is 20000, and the Selected Alternative is "Launch Product".

```
# Expected Value Calculation

# Launch Product
prob_high = 0.60
profit_high = 100000

prob_low = 0.40
loss_low = -30000

# Do Not Launch
profit_no_launch = 20000

# Calculate Expected Values
EV_launch = (
    prob_high * profit_high +
    prob_low * loss_low
)

EV_no_launch = profit_no_launch

# Display
print("Expected Value of Launching:", EV_launch)
print("Expected Value of Not Launching:", EV_no_launch)

# Select alternative
if EV_launch > EV_no_launch:
    print("Selected Alternative: Launch Product")
else:
    print("Selected Alternative: Do Not Launch")
s
```

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.

```
>>>
==== RESTART: C:/Users/info/OneDrive/Desktop/prac 6 decision/decision (a).py ====

==== RESTART: C:/Users/info/OneDrive/Desktop/prac 6 decision/DECISION(B).py ====
Expected Value of Launching: 48000.0
Expected Value of Not Launching: 20000
Selected Alternative: Launch Product
>>>
```